Arista Access Node WFD-ACC-31 Routing Protocol Flap Resolution SOP

Node: WFD-ACC-31 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_3292

Description

This Standard Operating Procedure (SOP) addresses the occurrence of routing protocol flap on the Arista Access Node, WFD-ACC-31, located at the Whitefield POP. Routing protocol flap refers to the repeated establishment and teardown of routing protocol sessions, typically characterized by BGP route churn, OSPF adjacency loss, or ISIS LSP flooding. This phenomenon can be attributed to various factors, including misconfigured routing policies, inadequate network design, or hardware and software issues. The Arista EOS software, running on the access node, utilizes a modular architecture that separates the control and data planes. The control plane is responsible for maintaining routing protocol sessions, while the data plane handles packet forwarding. A routing protocol flap on the WFD-ACC-31 node can be caused by issues within the control plane, such as BGP route reflection or OSPF area border router misconfiguration, or data plane problems, such as high CPU utilization or packet buffer exhaustion. Therefore, this SOP focuses on troubleshooting and resolving routing protocol flap issues on the Arista Access Node, WFD-ACC-31, at the Whitefield POP.

Pre-requisite Checks

- Physical Check: Verify power supply to the node is stable and not overloaded., Environmental Check: Check temperature and humidity levels within the data center.

- Link Status: Verify the status of all redundant links connected to the node.

Potential Root Causes

Cause	Probability
BGP route reflection misconfiguration	High
OSPF area border router misconfiguration	Medium
ISIS LSP flooding caused by hardware or software issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: Displayed BGP neighbor and session information., Description: This command provides an overview of BGP neighbor and session information, including the number of established and active connections, and the number of prefixes received and sent.

Command: show ospf neighbor, Expected Output: Displayed OSPF neighbor information, including state and dead time., Description: This command displays OSPF neighbor information, including state and dead time, to help identify any misconfigured OSPF area border routers.
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```
Command: show isis database, Expected Output: Displayed ISIS LSP information, including type and sequence number., Description: This command displays ISIS LSP information, including type and sequence number, to help diagnose LSP flooding caused by hardware or software issues.
```

Resolution Steps

Step 1: Step: Verify BGP route reflection configuration, Command: show ip bgp confederation, Description: Verify BGP route reflection configuration to ensure it is correctly configured and not causing routing protocol flap.

Step 2: Step: Check OSPF area border router configuration, Command: show ospf database, Description: Verify OSPF area border router configuration to ensure it is correctly configured and not causing routing protocol flap.

Step 3: Step: Clear ISIS LSP cache, Command: clear isis database, Description: Clear the ISIS LSP cache to resolve LSP flooding caused by hardware or software issues.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: No BGP neighbor or session issues., Description: Verify that there are no issues with BGP neighbor or session information.

Command: show ospf neighbor, Expected Output: All OSPF neighbors are in a stable state., Description: Verify that all OSPF neighbors are in a stable state.

Command: show isis database, Expected Output: No ISIS LSP flooding issues., Description: Verify that there are no ISIS LSP flooding issues.
```

Roll-back Procedures

- Step: Restore previous BGP configuration, Command: copy nvram:startup-config running-config, Description: Restore the previous BGP configuration to resolve any changes made during the troubleshooting process.

- Step: Revert OSPF area border router configuration, Command: copy nvram:startup-config running-config, Description: Revert the OSPF area border router configuration to its original state.

- Step: Rebuild ISIS LSP cache, Command: clear isis database, Description: Rebuild the ISIS LSP cache to resolve any changes made during the troubleshooting process.